

Yash Dave

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Education

Stanford University

June 2027

M.S. Computational and Mathematical Engineering, Data Science Track (3.8 GPA)

Coursework: Theoretical Statistics, AI Measurement Science, AI Reliability, Convex Optimization, Preference Learning, HCI

University of California, Berkeley

May 2025

B.A. Applied Math (Economics cluster) and Computer Science (Honors); Public Policy Minor (3.94 GPA)

Coursework: ML*, Optimization*, Computer Vision*, Algorithms, Linear Algebra, Probability, Statistical Inference *graduate-level

Honors: Phi Beta Kappa, Outstanding GSI Award, Kevin K. Gong Memorial Scholarship , Upsilon Pi Epsilon, Dean's List

Publications & Works in Progress

M. Hardy, A. Reuel, L. Zhang, J. M. Casabianca, S. Truong, **Y. Dave**, H. Lee, B. W. Domingue, S. Koyejo. *AI Cartography: Mapping the Latent Landscape of AI Benchmark Ecosystems* . ICML 2026.

B. Marks*, **Y. Dave***, Z. Wang, H. Chung, R. Patwa, S. Cha, M. Murphy, A. Strang. *Do Generalized-Gamma Scale Mixtures of Normals Fit Large Image Datasets?* . arXiv:2512.17038, under review. *equal contribution

Y. Dave, S. Truong, S. Wang, S. Koyejo. *The AI Evaluation Ecosystem*. In preparation.

Research Experience

Snowflake

Menlo Park, CA

Research Intern

June–August 2026

- Designed and shipped a user-in-the-loop method that mines sessions to jointly improve memory, skills, hooks, and MCP
- Built multi-turn memory evals and harness-agnostic benchmark infra for Claude Code, Codex, OpenCode, and OpenHands
- Turned real agent sessions into an internal database-engineering benchmark graded by a calibrated LLM judge

Stanford Trustworthy AI Research Lab

Stanford, CA

Research Assistant (under Prof. Sanmi Koyejo)

June 2025–present

- Building an expert-validated generative agent-based model of the AI eval ecosystem to study evaluator and policy design
- Exploring methods for sample-efficient prediction of agent reliability (e.g., cold-start IRT, TabPFN)

Strang Lab

Berkeley, CA

Research Assistant (under Prof. Alexander Strang)

January 2024–September 2025

- Developed a testing pipeline to evaluate realism of a Bayesian mixture model  for modeling quasi-sparse image data
- Created a performant library  for generating CDFs of prior distributions at scale, finding best-fit parameters, and visualization

Teaching

Course Assistant, CS 229 Machine Learning , Stanford

Winter 2026

Course Assistant, CS 283 Governing Artificial Intelligence , Stanford

Autumn 2025

Teaching Assistant, Data 140 Probability  and Data 375 Pedagogy , UC Berkeley

Fall 2024–Spring 2025

Lead Content TA, Data 100 Principles of Data Science , UC Berkeley

Spring 2023–Spring 2024

- Co-authored a public debugging guide  and contributed to a data science reference text 

Selected Projects

Reliability and Validity of Personalized AI Benchmarks

June 2026

- Built a 110-item personalized coding-agent benchmark rated by the user and a cross-lineage LLM judge panel; fit a hierarchical Bayesian Bradley-Terry-Davidson model with generalizability and decision studies to find the cheapest reliable design
- Showed LLM judges are reliable but not valid: they overstate win rates (0.80 vs. the user's 0.57), share correlated errors that do not average out, and give prediction-powered inference no label savings

Direct Preference Optimization applied to Financial Advising

December 2025

- Fine-tuned a 4B-parameter model using DPO to automate SEC Marketing Rule compliance review, reducing reasoning tokens by 50% while maintaining expert-level rewrite quality
- Built a privacy-preserving synthetic data pipeline generating 943 preference pairs from 40 expert annotations using a teacher-fixer-breaker framework to train compliant ad reviewers

Beliefs about Govt. Intervention in India

May 2025

- Developed a generalized linear regression model to study the relationship between local development and pro-intervention beliefs, controlling for factors like demographics, credit attribution, and government satisfaction